

CORRESPONDENCE

A Novel Spookyvirus Outbreak of Regional and International Concern

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In late September 2024, clinicians in Brisbane, Australia, observed a small cluster of pneumonia cases among elderly residents of a nursing home in Kelvin Grove. Two patients were hospitalized on Sept 26 with acute lower respiratory illness, both testing negative for common respiratory viruses. Over the next week, additional residents and caregivers were hospitalized, prompting the isolation of the facility. In parallel, a separate family cluster involving pediatric patients and their caregiver emerged in Fairfield, with clinical presentations consistent with atypical pneumonia^{1,2}.

By Oct 5, 2024, Queensland Health reported an increasing number of unexplained hospitalizations for respiratory illness in the Brisbane Metro area. A total of 70 laboratory-confirmed patients had been hospitalized by Oct 20, most of whom were negative for routine viral panels. Among these were elderly individuals from long-term care facilities, pediatric patients, healthcare workers, and community members.

On Oct 16, researchers at the Peter Doherty Institute identified the causative agent as a novel spooky virus, provisionally named Spooky-like Virus 2024 (SPOK-24). SPOK-24 is genetically related to related to the Spookyvirus genus (Paramyxovirus family) but appears to have significantly lower virulence and higher human-to-human transmissibility than other spookyviruses.

The virus was rapidly sequenced and publicly shared on GenBank. By Oct 29, the Doherty Institute had developed and released RT-PCR primers targeting the viral L gene, enabling the first wave of widespread laboratory confirmations. By that time, however, local transmission had been established, and signs of national spread were evident, with confirmed hospitalizations in Townsville, Rockhampton, and Sydney. Contact tracing became unmanageable in Brisbane by late October.

Simultaneously, confirmed international importations were reported in New Zealand, Papua New Guinea, Singapore, Japan, the USA, and several European countries³. Among the

first confirmed international cases was an Australian expatriate returning to New York City on Oct 25 who tested positive after developing flu-like symptoms in transit. The case was isolated promptly, with no onward transmission detected.

A prospective clinical cohort of 70 hospitalized patients^{1,2} revealed a clinical spectrum dominated by fever (94%), non-productive cough (69%), and myalgia or fatigue (44%). Dyspnea developed in 33%, with radiographic evidence of viral pneumonia in 87% of cases. Laboratory abnormalities included lymphopenia (49%) and elevated CRP (81%). Notably, four patients (5.7%) developed neurologic complications, including two cases of encephalitis confirmed by viral detection in cerebrospinal fluid. Thirteen patients (19%) required ICU admission, and six deaths (8.6%) were recorded by Oct 20. These data suggest lower severity than classical Hendra virus, but sufficient morbidity to strain clinical services in affected regions. Epidemiologic analysis of the first 70 cases showed a mean incubation period of 2 to 4 days and a basic reproductive number (R_0) ranging from 1.2 to 2.3. No significant differences in attack rates were observed by sex.

On Nov 12, 2024, the World Health Organization declared a Public Health Emergency of International Concern (PHEIC) due to growing international spread, sustained human-to-human transmission, and continued uncertainty around the virus's ecological reservoir and clinical progression.

Although SPOK-24 appears to cause milder disease than other spooky-like viruses, its potential for asymptomatic transmission, neurologic complications, and global dissemination in the absence of pharmaceutical countermeasures remains a concern. No licensed antivirals or vaccines exist for SPOK-24. Therapeutic options are being explored based on monoclonal antibody libraries developed for Nipah virus.

This outbreak highlights the fragile boundary between zoonotic spillover and international epidemic spread. It reinforces the need for rapid genomic surveillance, cross-disciplinary outbreak intelligence, and preparedness for respiratory viruses beyond influenza and coronaviruses. As clinical and epidemiologic understanding of SPOK-24 deepens, international collaboration will be essential to guide diagnostics, therapeutics, and policy.

References

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4. Huang C, Wang Y, Li X, et al. Clinical features of patients infected with 2019 novel coronavirus in Wuhan, China. *Lancet.* 2020;395(10223):497–506.

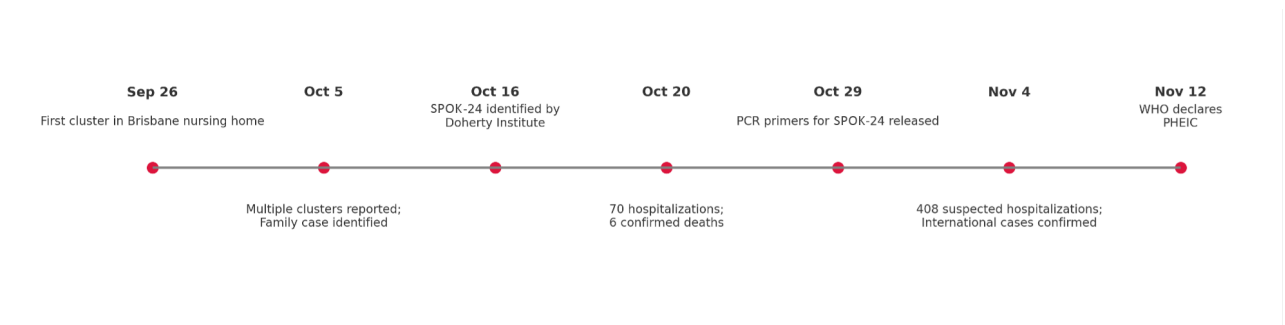


Figure 1: Timeline of key events.

Characteristic	SPOK-24	COVID-19 (early 2020)
Family (genus)	Paramyxoviridae (Spookyvirus)	Coronaviridae (Betacoronavirus)
Reservoir Host	Pteropus spp. (fruit bats)	Probable bats (via intermediate host)
Mean Incubation Period	2-4 days	4-7 days
Serial Interval	4-8 days	5-6 days
R_0	1.3-2.3	2.5 (initial estimate)
Common symptoms	Fever, cough, myalgia, fatigue	Fever, cough, fatigue, loss of smell
Primary Transmission	Droplet, contact	Droplet, aerosol
Severe manifestations	Pneumonia, encephalitis	Pneumonia, ARDS, Thrombotic events
Neurologic Complications	Yes (~5.7%)	Rare

Table 1: Comparative characteristics of Spooky-like Virus 2024 (SPOK-24) and SARS-CoV-2 (COVID-19)⁴ early findings, including epidemiological parameters, clinical manifestations, and origin.